A 13-variable natural equation: source-only candidate

Nonnegative integer unknowns; 87 binary-operation abbreviations; exact degree 60 at the fixed ports.

The 13 unknowns below are distinct nonnegative integers:
`argJ, argFuel, argAlpha, argPellF, argIndex, argTime, argW`
`argScale, argOrdinate, argEta, argZeta, argAuxY, argGamma`
Each monospaced G-name below is an abbreviation, not an unknown. The fixed integers β, K, μ, ω are defined after the equation.

$\texttt{G001} = \texttt{argFuel} + 1$
$\texttt{G002} = \texttt{argAlpha} + 1$
$\texttt{G003} = \texttt{argIndex} + 1$
$\texttt{G004} = \texttt{argTime} + 1$
$\texttt{G005} = \texttt{argScale} + 1$
$\texttt{G006} = \texttt{argEta} + 1$
$\texttt{G007} = \texttt{argJ} \cdot \beta$
$\texttt{G008} = 1 + \texttt{G007}$
$\texttt{G009} = \texttt{G008} \cdot \texttt{argW}$
$\texttt{G010} = \texttt{G008} \cdot \texttt{G008}$
$\texttt{G011} = \texttt{G008} \cdot \texttt{G010}$
$\texttt{G012} = \texttt{G011} \cdot \texttt{G005}$
$\texttt{G013} = \texttt{G012} \cdot \texttt{G010}$
$\texttt{G014} = \texttt{G008} - \texttt{G001}$
$\texttt{G015} = \texttt{G014} - \texttt{G002}$
$\texttt{G016} = \texttt{G009} \cdot \texttt{G012}$
$\texttt{G017} = \texttt{G006} + \texttt{argZeta}$
$\texttt{G018} = \texttt{G017} + 1$
$\texttt{G019} = \texttt{G012} \cdot \texttt{G018}$
$\texttt{G020} = \texttt{G006} + \texttt{G019}$
$\texttt{G021} = \texttt{G020} \cdot \texttt{G020}$
$\texttt{G022} = \texttt{G016} + \texttt{G012}$
$\texttt{G023} = 4 \cdot \texttt{G022}$
$\texttt{G024} = 3 + \texttt{G023}$
$\texttt{G025} = \texttt{G022} \cdot \texttt{G022}$
$\texttt{G026} = \texttt{G024} + \texttt{G025}$
$\texttt{G027} = \texttt{G022} \cdot \texttt{G020}$
$\texttt{G028} = \texttt{G009} + \texttt{G027}$
$\texttt{G029} = \texttt{G016} \cdot \texttt{G019}$
$\texttt{G030} = \texttt{G009} + K$
$\texttt{G031} = \texttt{G015} \cdot \texttt{G030}$
$\texttt{G032} = \texttt{G031} - \texttt{G001}$
$\texttt{G033} = \texttt{G029} + \texttt{G018}$
$\texttt{G034} = \texttt{G033} + \texttt{argOrdinate}$
$\texttt{G035} = \texttt{G018} \cdot \texttt{G032}$
$\texttt{G036} = \texttt{G034} - \texttt{G035}$
$\texttt{G037} = \texttt{G007} \cdot \texttt{G036}$
$\texttt{G038} = \texttt{G037} - \texttt{G032}$
$\texttt{G039} = \texttt{G038} \cdot \texttt{G038}$
$\texttt{G040} = \texttt{G007} \cdot \texttt{G007}$
$\texttt{G041} = \texttt{G029} \cdot \texttt{G033}$
$\texttt{G042} = 1 + \texttt{G041}$
$\texttt{G043} = \texttt{G040} \cdot \texttt{G042}$
$\texttt{G044} = \texttt{G039} - \texttt{G043}$
$\texttt{G045} = \texttt{G014} \cdot \texttt{G007}$
$\texttt{G046} = \texttt{G002} + \texttt{G045}$
$\texttt{G047} = \texttt{G010} - 1$
$\texttt{G048} = \texttt{G046} \cdot \texttt{G047}$
$\texttt{G049} = \omega \cdot \texttt{G008}$
$\texttt{G050} = \mu + \texttt{G049}$
$\texttt{G051} = \texttt{argJ} \cdot \texttt{G050}$
$\texttt{G052} = \texttt{G048} + \texttt{G051}$
$\texttt{G053} = \texttt{G017} - \texttt{G052}$
$\texttt{G054} = \texttt{G013} \cdot \texttt{argGamma}$
$\texttt{G055} = \texttt{G054} - \texttt{G053}$
$\texttt{G056} = \texttt{G013} \cdot \texttt{G028}$
$\texttt{G057} = \texttt{G024} \cdot \texttt{G055}$
$\texttt{G058} = \texttt{G056} + \texttt{G057}$
$\texttt{G059} = \texttt{G058} \cdot \texttt{G058}$
$\texttt{G060} = \texttt{G013} \cdot \texttt{G013}$
$\texttt{G061} = \texttt{G026} \cdot \texttt{G021}$
$\texttt{G062} = 1 + \texttt{G061}$
$\texttt{G063} = \texttt{G060} \cdot \texttt{G062}$
$\texttt{G064} = \texttt{G059} - \texttt{G063}$
$\texttt{G065} = \texttt{argPellF} \cdot \texttt{argPellF}$
$\texttt{G066} = \texttt{G065} - 1$
$\texttt{G067} = \texttt{G026} \cdot \texttt{G066}$
$\texttt{G068} = \texttt{G021} \cdot \texttt{G003}$
$\texttt{G069} = \texttt{G068} \cdot \texttt{G068}$
$\texttt{G070} = \texttt{G069} - \texttt{G067}$
$\texttt{G071} = \texttt{G004} \cdot \texttt{G020}$
$\texttt{G072} = \texttt{G071} - \texttt{G052}$
$\texttt{G073} = \texttt{G065} \cdot \texttt{G072}$
$\texttt{G074} = \texttt{G073} - \texttt{G020}$
$\texttt{G075} = \texttt{argAuxY} \cdot \texttt{argAuxY}$
$\texttt{G076} = \texttt{G074} \cdot \texttt{G074}$
$\texttt{G077} = \texttt{G075} - \texttt{G076}$
$\texttt{G078} = \texttt{G067} \cdot \texttt{G077}$
$\texttt{G079} = \texttt{G075} - \texttt{G078}$
$\texttt{G080} = \texttt{G044} \cdot \texttt{G044}$
$\texttt{G081} = \texttt{G064} \cdot \texttt{G064}$
$\texttt{G082} = \texttt{G079} - 1$
$\texttt{G083} = \texttt{G082} \cdot \texttt{G082}$
$\texttt{G084} = \texttt{G070} \cdot \texttt{G070}$
$\texttt{G085} = \texttt{G080} + \texttt{G081}$
$\texttt{G086} = \texttt{G083} + \texttt{G084}$
$\texttt{G087} = \texttt{G085} + \texttt{G086}$

The single equation is $\boxed{\texttt{G087} = 0}$.
The zero-map to the positive parent is an external standard-integer argument; its nonstandard model transfer and iPRA endpoint remain open.

The four fixed integers. All numerals, including the complete seed below, use ordinary base-10 notation. Put $\rho = 2^{67}$, $\beta = \rho^{8807} - 1$, and let $d_j = \lfloor P/10^{16j} \rfloor \bmod 10^{16}$ for $0 \le j < 943$. Let the ordered list $(\lambda_0, \ldots, \lambda_{40})$ be

$$\begin{gathered}1023, 1024, 1025, 1026, 1027, 1028, 1029, 1030, 1013, 1014, 1015, 1016, 1017, 1018,\\ 1019, 1020, 1021, 1003, 1004, 1005, 1006, 1007, 1008, 1009, 1010, 983, 984, 985,\\ 986, 987, 988, 989, 990, 991, 1031, 1022, 1011, 992, 1095, 2195, 3296\end{gathered}$$

Define

$$\begin{aligned}K = {} & \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849}\\ & + \sum_{j=0}^{942} d_j (\rho^{4393-j} + \rho^{5495-j})\\ & + \sum_{j=0}^{40} 2^{10+j} (\rho^{4393-\lambda_j} + \rho^{5495-\lambda_j}).\end{aligned}$$

For integer intervals $[a, b] = \{a, a+1, \ldots, b\}$, let

$$\begin{aligned}S = (& [0, 942] \cup [983, 992] \cup [1003, 1011] \cup [1013, 1031]\\ & \cup \{1063, 1064, 1095, 2195, 3296\}) \setminus \{1\}.\end{aligned}$$

Put $\mu = \beta - \sum_{k\in S} \rho^k - 2\rho^{1063}$ and $\omega = \beta + H$, where

$$H = 4 + \sum_{k=1013}^{1022} \rho^k + \sum_{k=1065}^{1073} \rho^k + \rho^{2195} + \rho^{3296}$$

$$+ (2^{51} - 2)(\rho^{4393} + \rho^{5495}) + (2^{65} - 1) \sum_{k=8792}^{8804} \rho^k + (2^{19} - 1)\rho^{8805}.$$

Complete decimal integer *P*. Concatenate these lines in order, without spaces or line breaks:

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The compact fixed-port recipe is distinct from the current-2011 equation. The external parent review is for standard integers and does not establish an iPRA endpoint for this expression.